Kyle Rassweiler

Engineering Specialist and Software Developer

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Profiles

rassweiler

kylerassweiler

KyleRassweiler

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Skills

Web Technologies

HTML, JS, Typescript, PHP, React, Django, SCSS, Redux, Angular

Development

C++, Python, Git, Assembly, Qt

Databases

MySQL, MariaDB, MongoDB, SQLite, PostgreSQL

OS

Linux, Qemu, KVM, Proxmox, Docker

PLC

Mitsubishi, Toyopuc, Studio 5000, Connected Components, Omron, Siemens

Robots

Kawasaki, Nachi, Panasonic, Yaskawa, Motoman

Vision

Cortex, Robeye, Keyence, Omron

CAD

FreeCAD, Kicad, AutoCAD, Catia, Delmia

Summary

I'm an Engineering specialist with experience in both software and hardware development with schooling focused on embedded development and robotics. I'm a self starter and quick to learn new technologies.

Experience

Canopy Growth — 2023-Present

Controls & Automation — Smiths Falls

- Implemented additive manufacturing process and taught maintenance/ engineering groups CAD systems, saving the company $160000 in part purchases within a year.
- Implemented an embedded system I designed to count products and pacing while uploading the data to SharePoint, saving the company $19500 per unit compared to licenses for existing solutions.
- Perform Software Development, Commissioning, HMI design, PLC programming (AB, Siemens, Omron), SCADA system design, Building Automation (Ecostruxure), CAD design, Project Management, Additive Part manufacturing, Electrical Schematics/ Diagrams, Work order management (Fiix), Instrumentation & Calibration, Vision Systems (Keyence, Cognex)

Toyota (TMNA) — 2017-2023

Engineering Specialist — Cambridge, ON

https://www.toyota.com/usa/operations/map/tcal

- Managed project implementation during non production time, ensuring no issues affect production and quality.
- Implemented a new teach method that saved 75% of equipment recovery time, using a 4 point before-after comparison.
- Setup production lines in 7 plants across North America for: Rav4, Camery, Hilander, Lexus NX, and Lexus RX. Production earned roughly 89 billion USD in 2022 across North America
- Perform Software Development (VBA, JS), Commissioning, HMI design, PLC programming (GXWorks, Toyopuc), 3D factory design/ CAD (Delmia, Catia), Robot programming, Robotic vision (Robeye, Cortex), Project management, Vision Systems (Keyence, Omron)

FIO Automotive — 2014-2017

Robotics Technician — Stratford, ON

- Developed a proof of concept SCADA system using Angular, MongoDB, and Node.js. The system allowed management to track downtime and target repeat issues during a 24/7 production period.
- Setup networking for all production lines in order to implement my proof of concept and collect data to a MSSQL database.
- Software Development (VBA, JS, React), Commissioning, HMI design, PLC programming (GXWorks, Toyopuc), Robot programming, Project management

Certifications

Responsive Web Development
FreeCodeCamp
June 2022
Certificate

JavaScript Algorithms and Data Structures
FreeCodeCamp
July 2022
Certificate

Front End Development Libraries
FreeCodeCamp
February 2023
Certificate

Projects

Networked Counter **2025-Present**
An embedded product counter with networked exporting.
https://github.com/rassweiler/pi-networked-counter

3d printed counter using Pi5, python, pyqt6, and sqlite that can export to folders or sharepoint. Used to count items travelling across one or two sets of infra red sensors

Rasberry Pi, Python, Qt, Sqlite, SharePoint, Linux

Krello **February 2023**
Kanban web app
https://krello.kylerassweiler.ca/

A simple kanban web app made using Nextjs, React, tRPC, NextAuth, Prisma, and Typescript. The app is hosted on Vercel and Planetscale

TypeScript, React, tRPC, Nextjs, NextAuth, Prisma, Vercel, PlanetScale

Education

Algonquin College **2010-2012**
Electro Mechanical Engineering Diploma
https://www.algonquincollege.com/sat/program/electro-mechanical-engineering-technician-robotics/

Embedded development, plc programming, circuit and pcb design, motor controls, industrial robotics, microcontrollers

Algonquin College **2007-2010**
Computer Animation Advanced Diploma
https://www.algonquincollege.com/mediaanddesign/program/animation/

Traditional animation, 3d animation, Rigging, Modelling, Python scripting.

Algonquin College **2006-2007**
Drawing Founadations Certificate
https://www.algonquincollege.com/mediaanddesign/program/drawing-foundations-for-animation-and-illustration/